

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Artificial Intelligence and Data Science**  
**ASSESSMENT TOOL 2 – Technical Presentation**

|                         |                                                                                                                                                          |                      |                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------|
| <b>Course Code:</b>     | DSA03                                                                                                                                                    | <b>Course Title:</b> | Natural Language Processing                |
| <b>CO Assessed:</b>     | CO1 – Imitation (L1)                                                                                                                                     | <b>Assessment:</b>   | Assessment Tool 2 – Technical Presentation |
| <b>Weightage:</b>       | 30% of CIA                                                                                                                                               |                      |                                            |
| <b>CO1 Statement:</b>   | Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3) |                      |                                            |
| <b>Student Name:</b>    | B. Rajesh                                                                                                                                                | <b>Reg. No.:</b>     | 192472165                                  |
| <b>Section / Batch:</b> | CSE--AI                                                                                                                                                  | <b>Date:</b>         |                                            |

**Code of Conduct**

*I B. Rajesh(192472165) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*  
Signature of the Student: B.Rajesh

**Instructions**

- Suggested Team Member Roles (3 Members)
  - Member 1: Theory, Concepts, and Literature Review
  - Member 2: Algorithm Design, Models, and Technical Analysis •
  - Member 3: Demonstration, Case Study, Applications, and Conclusion •
- GPS photo must submit.
- Duration: 30 minutes.

| Section / Topic                                                                 | Focus                                     | Q Nos |
|---------------------------------------------------------------------------------|-------------------------------------------|-------|
| Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP | NLP Models, Language Processing Evolution | TP-01 |

|                                                                               |                                                 |       |
|-------------------------------------------------------------------------------|-------------------------------------------------|-------|
| Comparative Study of NLP Models and Algorithms                                | Model Analysis, Algorithm Comparison            | TP-02 |
| Regular Expressions for Real-World Text Processing                            | Pattern Matching, Information Extraction        | TP-03 |
| Designing Efficient Regular Expression Patterns for Information Extraction    | Text Mining, Data Extraction Techniques         | TP-04 |
| Finite State Automata in Natural Language Processing                          | State Transitions, Language Recognition         | TP-05 |
| Building a Finite State Automaton for Token Recognition                       | Lexical Analysis, Token Validation              | TP-06 |
| Morphology and Word Formation in Natural Languages                            | Morpheme Analysis, Word Structure Processing    | TP-07 |
| Inflectional Morphology: Understanding Grammatical Variations                 | Grammatical Analysis, Word Normalization        | TP-08 |
| Derivational Morphology and Vocabulary Expansion                              | Word Generation, Semantic Transformation        | TP-09 |
| Comparative Analysis of Inflectional and Derivational Morphology              | Morphological Classification, Language Analysis | TP-10 |
| Finite-State Morphological Parsing Techniques                                 | Morphological Parsing, Finite-State Processing  | TP-11 |
| Morphological Analysis for Indian Languages                                   | Language-Specific Morphology, NLP Challenges    | TP-12 |
| Porter Stemmer: Design, Working Mechanism, and Applications                   | Stemming Algorithms, Text Preprocessing         | TP-13 |
| Stemming vs Morphological Parsing: A Comparative Study                        | Word Reduction, Morphological Processing        | TP-14 |
| Integrated NLP Pipeline: From Regular Expressions to Morphological Processing | End-to-End Language Processing Pipeline         | TP-15 |

## Annexure A – Technical Presentation

| Team No. | Technical Presentation Title                                                    | Description                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP | Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.              |
| 2        | Comparative Study of NLP Models and Algorithms                                  | Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.                             |
| 3        | Regular Expressions for Real-World Text Processing                              | Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.                 |
| 4        | Designing Efficient Regular Expression Patterns for Information Extraction      | Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.                    |
| 5        | Finite State Automata in Natural Language Processing                            | Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.                              |
| 6        | Building a Finite State Automaton for Token Recognition                         | Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.                                    |
| 7        | Morphology and Word Formation in Natural Languages                              | Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.                                             |
| 8        | Inflectional Morphology: Understanding Grammatical Variations                   | Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.                                               |
| 9        | Derivational Morphology and Vocabulary Expansion                                | Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.                                    |
| 10       | Comparative Analysis of Inflectional and Derivational Morphology                | Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.                                                   |
| 11       | Finite-State Morphological Parsing Techniques                                   | Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.                                   |
| 12       | Morphological Analysis for Indian Languages                                     | Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.        |
| 13       | Porter Stemmer: Design, Working Mechanism, and Applications                     | Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.                                      |
| 14       | Stemming vs Morphological Parsing: A Comparative Study                          | Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.                                         |
| 15       | Integrated NLP Pipeline: From Regular Expressions to Morphological Processing   | Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing. |

## Rubrics for Calculation

| Technical Presentation     |           |                                                                    |                                                                |                                                            |                                                                     |
|----------------------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weightage | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                       | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30        | Highly accurate, indepth, and insightful technical explanation     | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth   | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25        | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak     | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20        | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation         | Minimal or incorrect use of tools and data; poor visuals            |
| Communication              | 15        | Highly engaging, confident, and audience-focused delivery          | Clear, organized, and professional communication               | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides               |
| Professionalism            | 10        | Exemplary professionalism, leadership, and ethical awareness       | Professional conduct with effective team collaboration         | Basic professionalism ; uneven team participation          | Lacks professionalism ; minimal contribution or ethical awareness   |

| Team No. / Criteria Level | Technical Content Accuracy | Problem Identification | Use of Tools | Communication | Professionalism | Total % |
|---------------------------|----------------------------|------------------------|--------------|---------------|-----------------|---------|
| 6                         | 22                         | 19                     | 20           | 15            | 10              | 86      |

| Faculty In-charge    | Course Coordinator | Program Director |
|----------------------|--------------------|------------------|
| Dr.T.Kumaragurubaran |                    |                  |
| Signature: _____     | Signature: _____   | Signature: _____ |
| Date: _____          | Date: _____        | Date: _____      |



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# Regular Expressions-Regex

**PRESENTED BY**

B. Rajesh(192472165)



# INTRODUCTION

- Definition:** A Regular Expression is a compact sequence of characters that defines a specific text search pattern.
- Purpose:** It is primarily used to search, match, and manipulate text strings based on complex rules.
- Mechanism:** It combines literal characters and special symbols to find exact matches or structural patterns.



## USES

- **Data Validation:** Checking if inputs like emails, phone numbers, and passwords match required formats.
- **Search and Replace:** Finding specific text patterns in a document and automatically updating them.
- **Data Extraction:** Parsing raw text or web pages to isolate and extract specific keywords or values.
- **Log Analysis:** Scanning system logs to quickly identify error codes.





## ADVANTAGES

- **High Efficiency:** A single short regex string can replace dozens of lines of traditional logic code.
- **Fast Execution:** Built-in engines are highly optimized to process massive text datasets incredibly quickly.
- **Universal Standard:** The core syntax remains virtually identical across different programming platform.

# DISADVANTAGES

- Steep Learning Curve:** The syntax is highly cryptic, making it difficult for beginners to read and memorize.
- Hard to Maintain:** Complex expressions are tough to debug and modify after they are written.
- Performance Risks:** Poorly designed patterns can cause systems to freeze due to excessive backtracking.
- No Context Awareness:** It cannot reliably parse deeply nested.



## CONCLUSION

- Essential Tool:** Regex is an invaluable asset for efficient text processing in modern software development.
- Readability Balance:** Developers must use it carefully to ensure the code remains easy for others to understand.

- Final Verdict:** When used correctly, it drastically simplifies text manipulation and saves significant development time.

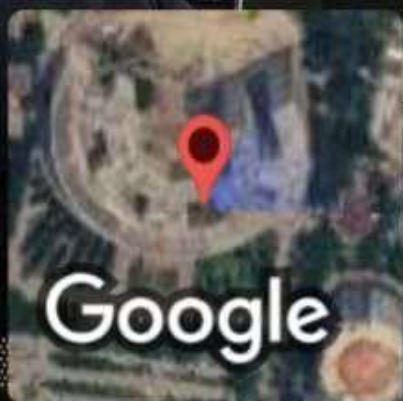
A vibrant watercolor illustration of a butterfly, centered on the page. The butterfly's wings are painted with a variety of colors including deep purple, bright pink, fiery orange, and cool blue, with soft, blended edges. Small, dark brown dots are scattered around the butterfly, suggesting motion or pollen. The background is a solid, light peach color.

Thank  
you

Three state morphological analysis  
- keeps track of some  
- change morphology efficiently  
- may not use for  
- custom word  
- ...

### ADVANTAGES

- **High Efficiency:** A single short regex string can replace dozens of lines of traditional logic code.
- **Fast Execution:** Built-in engines are highly optimized to process massive text datasets incredibly quickly.
- **Universal Standard:** The core syntax remains virtually identical across different programming platforms.
- **Compact Code:** It minimizes boilerplate code, keeping scripts clean and straightforward.



Mevalurkuppam, Tamil Nadu,  
India 🇮🇳

Cse & It Department, Mevalurkuppam, Tamil Nadu  
602105, India

Lat 13.026026° Long 80.016736°

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